VehiSights Updated Database Data Dictionary

Current Supabase Schema for Capstone Documentation

This document defines the current VehiSights database structure used by the Laravel web application, the manual traffic-counting workflow, scheduled camera counting, reporting, prediction, Google account verification, and authenticated API access. The definitions reflect the live Supabase PostgreSQL schema reviewed on November 5, 2026.

Scope note. Nullable fields may contain no value. JSON fields store structured vehicle-count or configuration data. The live PostgreSQL schema uses timestamp values without an attached time-zone designation.

Table 1 Users

users

Stores local and Google-authenticated user accounts.

Field Name	Data Type	Description
id	bigint	Unique identifier of the user.
name	varchar(255), nullable	Complete display name of the user.
email	varchar(255)	Unique email address used for authentication.
email_verified_at	timestamp, nullable	Date and time the email address was verified.
password	varchar(255), nullable	Hashed password; may be null for Google-only accounts.
remember_token	varchar(100), nullable	Token used by Laravel's Remember Me authentication.
department	varchar(255)	Department or office of the user; defaults to City Planning.
is_active	boolean	Indicates whether the account is allowed to access the system.
role	varchar(20), nullable	Access role such as viewer, analyst, admin, or super_admin.
first_name	varchar(255), nullable	First name of the user.
last_name	varchar(255), nullable	Last name of the user.
google_verified_at	timestamp, nullable	Date and time Google account verification was completed.
created_at	timestamp, nullable	Date and time the account was created.
updated_at	timestamp, nullable	Date and time the account was last updated.

Table 2 Social Accounts

social_accounts

Connects a system user to an external authentication provider.

Field Name	Data Type	Description
id	bigint	Unique identifier of the social account record.
user_id	bigint	References users.id; deleted with the parent user.
provider	varchar(32)	Authentication provider, such as google.

Field Name	Data Type	Description
provider_user_id	varchar(255)	Unique account identifier supplied by the provider.
created_at	timestamp, nullable	Date and time the connection was created.
updated_at	timestamp, nullable	Date and time the connection was updated.

Table 3 Email Verification Codes

email_verification_codes

Stores temporary verification codes for Google-authenticated accounts.

Field Name	Data Type	Description
id	bigint	Unique identifier of the verification record.
user_id	bigint	Unique reference to users.id; deleted with the parent user.
code_hash	varchar(255)	Secure hash of the verification code.
expires_at	timestamp	Date and time the verification code expires.
attempts	smallint	Number of unsuccessful verification attempts.
last_sent_at	timestamp	Date and time the most recent code was sent.
created_at	timestamp, nullable	Date and time the record was created.
updated_at	timestamp, nullable	Date and time the record was updated.

Table 4 Personal Access Tokens

personal_access_tokens

Stores Laravel Sanctum API tokens used by external applications such as TrafficCounter for macOS.

Field Name	Data Type	Description
id	bigint	Unique identifier of the API token.
tokenable_type	varchar(255)	Model type that owns the token.
tokenable_id	bigint	Identifier of the user or model that owns the token.
name	varchar(255)	Descriptive name of the token or client device.
token	varchar(64)	Securely hashed API token.
abilities	text, nullable	Permissions assigned to the token.
last_used_at	timestamp, nullable	Date and time the token was last used.
expires_at	timestamp, nullable	Optional expiration date and time.
created_at	timestamp, nullable	Date and time the token was created.
updated_at	timestamp, nullable	Date and time the token was updated.

Table 5 Cameras

cameras

Stores cameras registered for scheduled automatic vehicle counting.

Field Name	Data Type	Description
id	bigint	Unique identifier of the camera.
name	varchar(255)	Descriptive camera name.
ip_address	varchar(255)	Unique IP address of the camera.
location	varchar(255)	Physical location of the camera.
stream_url	varchar(255), nullable	Optional RTSP or HTTP video stream URL.
status	varchar(255)	Operational status, normally online or offline.
last_detection_at	timestamp, nullable	Date and time of the most recent vehicle detection.
detection_sensitivity	integer	Detection sensitivity setting; defaults to 50.
roi	json, nullable	Region-of-interest configuration for vehicle detection.
created_by	bigint	References users.id; user who registered the camera.
created_at	timestamp, nullable	Date and time the camera was registered.
updated_at	timestamp, nullable	Date and time the camera was updated.

Table 6 Sessions

sessions

Stores the main record for a manually counted traffic session.

Field Name	Data Type	Description
id	bigint	Unique identifier of the manual session.
session_code	varchar(255)	Unique code assigned to the session.
road_name	varchar(255)	Road or station being monitored.
barangay	varchar(255), nullable	Barangay where the road is located.
city	varchar(255)	City or municipality where the road is located.
direction	varchar(255)	Direction of traffic flow.
lanes	integer	Number of lanes included in the count.
has_median	boolean	Indicates whether the road has a median divider.
base_capacity_per_lane	integer	Road capacity per lane in PCU per hour.
start_time	timestamp	Date and time the manual session started.
end_time	timestamp, nullable	Date and time the manual session ended.
status	varchar(255)	Session state, such as active or ended.
created_by	bigint	References users.id; user who performed the count.
created_at	timestamp, nullable	Date and time the session was created.

Field Name	Data Type	Description
updated_at	timestamp, nullable	Date and time the session was updated.

Table 7 Manual Counts

manual_counts

Stores vehicle counts and calculated results for manual counting sessions.

Field Name	Data Type	Description
id	bigint	Unique identifier of the manual count record.
session_id	bigint, nullable	Unique reference to sessions.id; deleted with the parent session.
road_name	varchar(255)	Road or station where the count was conducted.
barangay	varchar(255), nullable	Barangay where the road is located.
city	varchar(255)	City or municipality where the road is located.
direction_from	varchar(255), nullable	Starting point of traffic movement.
direction_to	varchar(255), nullable	Destination of traffic movement.
count_date	date	Date the traffic count was conducted.
hour_start	time	Starting time of the count interval.
hour_end	time	Ending time of the count interval.
has_median	boolean	Indicates whether the road has a median.
lanes	integer	Number of lanes included in the count.
counts	json	Vehicle counts for the eight vehicle classifications.
pcu_totals	json	PCU-adjusted totals for each vehicle classification.
total_pcu	double precision	Combined Passenger Car Unit total.
vc_ratio	double precision	Computed volume-to-capacity ratio.
los	char(1)	Level of Service classification from A to F.
created_by	bigint	References users.id; user who recorded the count.
created_at	timestamp, nullable	Date and time the count was saved.
updated_at	timestamp, nullable	Date and time the count was updated.

Table 8 Session Details

session_details

Legacy detail table retained for compatibility with older manual session records.

Field Name	Data Type	Description
id	bigint	Unique identifier of the detail record.
session_id	bigint	Unique reference to sessions.id; deleted with the parent session.

Field Name	Data Type	Description
total_pcu	numeric(10,2)	Total Passenger Car Unit volume.
vc_ratio	numeric(5,2)	Volume-to-capacity ratio.
los	char(1)	Level of Service classification from A to F.
counts	json	Vehicle counts per classification.
pcu_totals	json	PCU-adjusted values per vehicle classification.
source	varchar(20)	Source of the count; defaults to manual.
created_at	timestamp, nullable	Date and time the record was created.
updated_at	timestamp, nullable	Date and time the record was updated.

Table 9 Schedules

schedules

Stores configurations for scheduled automatic vehicle counting.

Field Name	Data Type	Description
id	bigint	Unique identifier of the schedule.
schedule_name	varchar(255)	Descriptive name of the scheduled count.
camera_id	bigint	References cameras.id.
start_date	date	Date the schedule begins.
end_date	date, nullable	Optional date the schedule ends.
start_time	time	Daily start time of automatic counting.
end_time	time	Daily end time of automatic counting.
recurrence	varchar(255)	Recurrence mode such as once, daily, weekly, or monthly.
recurrence_days	json, nullable	Selected days used by recurring schedules.
is_active	boolean	Indicates whether the schedule is enabled.
created_by	bigint	References users.id; user who created the schedule.
created_at	timestamp, nullable	Date and time the schedule was created.
updated_at	timestamp, nullable	Date and time the schedule was updated.
road_name	varchar(255), nullable	Road or station monitored by the schedule.
barangay	varchar(255), nullable	Barangay where the camera is assigned.
city	varchar(255), nullable	City or municipality of the scheduled count.
direction_from	varchar(255), nullable	Starting direction of traffic flow.
direction_to	varchar(255), nullable	Destination direction of traffic flow.
lanes	integer, nullable	Number of lanes monitored.
has_median	boolean, nullable	Indicates whether the road has a median.

Field Name	Data Type	Description
base_capacity_per_lane	integer, nullable	Capacity of each lane in PCU per hour.

Table 10 Schedule Logs

schedule_logs

Stores the execution history of automatic counting schedules.

Field Name	Data Type	Description
id	bigint	Unique identifier of the schedule log.
schedule_id	bigint	References schedules.id; deleted with the parent schedule.
auto_session_id	bigint, nullable	References auto_sessions.id; set to null if that session is deleted.
status	varchar(255)	Execution state such as pending, running, success, or failed.
error_message	text, nullable	Error details when schedule execution fails.
started_at	timestamp, nullable	Date and time automatic execution started.
completed_at	timestamp, nullable	Date and time automatic execution completed.
created_at	timestamp, nullable	Date and time the log was created.
updated_at	timestamp, nullable	Date and time the log was updated.

Table 11 Auto Sessions

auto_sessions

Stores the main record and configuration snapshot for an automatic camera counting session.

Field Name	Data Type	Description
id	bigint	Unique identifier of the automatic session.
session_code	varchar(255)	Unique automatic session code.
camera_name	varchar(255)	Snapshot of the camera name during the session.
ip_address	varchar(255)	Snapshot of the camera IP address.
location	varchar(255)	Snapshot of the camera location.
start_time	timestamp	Date and time automatic counting started.
end_time	timestamp, nullable	Date and time automatic counting ended.
status	varchar(255)	Session state such as active or ended.
created_by	bigint	References users.id; user responsible for the session.
created_at	timestamp, nullable	Date and time the session was created.
updated_at	timestamp, nullable	Date and time the session was updated.
schedule_id	bigint, nullable	References schedules.id; set to null if the schedule is deleted.
camera_id	bigint, nullable	References cameras.id; set to null if the camera is deleted.

Field Name	Data Type	Description
road_name	varchar(255), nullable	Road or station monitored during the session.
barangay	varchar(255), nullable	Barangay of the monitored road.
city	varchar(255), nullable	City or municipality of the monitored road.
direction	varchar(255), nullable	Direction of traffic flow.
lanes	integer, nullable	Number of lanes monitored.
has_median	boolean, nullable	Indicates whether the road has a median.
base_capacity_per_lane	integer, nullable	Road capacity per lane in PCU per hour.

Table 12 Auto Session Details

auto_session_details

Stores the vehicle counts and calculated results of an automatic session.

Field Name	Data Type	Description
id	bigint	Unique identifier of the automatic detail record.
auto_session_id	bigint	Unique reference to auto_sessions.id; deleted with the parent session.
total_pcu	numeric(10,2)	Total Passenger Car Unit volume.
vc_ratio	numeric(5,2)	Computed volume-to-capacity ratio.
los	char(1)	Level of Service classification from A to F.
counts	json	Automatically detected counts per vehicle classification.
pcu_totals	json	PCU-adjusted total per vehicle classification.
source	varchar(20)	Source identifier; defaults to auto.
created_at	timestamp, nullable	Date and time the result was created.
updated_at	timestamp, nullable	Date and time the result was updated.

Table 13 Traffic History

traffic_history

Stores normalized historical traffic data for dashboards, reports, and prediction analysis.

Field Name	Data Type	Description
id	bigint	Unique identifier of the historical record.
camera_id	bigint, nullable	References cameras.id; set to null if the camera is deleted.
road_name	varchar(255)	Road or station represented by the record.
direction	varchar(255), nullable	Traffic flow direction.
hour_timestamp	timestamp	Date and time represented by the traffic record.
total_pcu	numeric(10,2)	Total Passenger Car Unit volume.

Field Name	Data Type	Description
vc_ratio	numeric(5,2)	Volume-to-capacity ratio.
los	char(1)	Level of Service classification from A to F.
counts	json	Vehicle counts per classification.
source	varchar(255)	Record source, either manual or auto.
created_at	timestamp, nullable	Date and time the history record was created.
updated_at	timestamp, nullable	Date and time the record was updated.

Table 14 Predictions

predictions

Stores generated traffic predictions and comparison values from observed traffic data.

Field Name	Data Type	Description
id	bigint	Unique identifier of the prediction.
camera_id	bigint, nullable	References cameras.id when the prediction is camera-specific.
road_name	varchar(255), nullable	Road or station covered by the prediction.
direction	varchar(255), nullable	Traffic direction represented by the prediction.
prediction_type	varchar(255)	Prediction category or model type.
prediction_date	date	Date for which the prediction applies.
prediction_hour	integer, nullable	Hour of the day represented by the prediction.
predicted_pcu	numeric(10,2)	Predicted Passenger Car Unit volume.
predicted_vc_ratio	numeric(5,2)	Predicted volume-to-capacity ratio.
predicted_los	char(1)	Predicted Level of Service from A to F.
confidence_score	numeric(5,2)	Confidence score assigned to the prediction.
actual_pcu	numeric(10,2), nullable	Actual PCU recorded after the predicted period.
model_version	varchar(255)	Version of the prediction model.
created_at	timestamp, nullable	Date and time the prediction was generated.
updated_at	timestamp, nullable	Date and time the prediction was updated.

Laravel Support Tables

The following tables support Laravel framework operations. They may be placed in an appendix when the capstone manuscript focuses only on the application's business data.

Table 15 Password Reset Tokens

password_reset_tokens

Stores temporary password-reset credentials.

Field Name	Data Type	Description
email	varchar(255)	Email address requesting a password reset.
token	varchar(255)	Secure password-reset token.
created_at	timestamp, nullable	Date and time the token was generated.

Table 16 Failed Jobs

failed_jobs

Stores background queue jobs that could not be completed.

Field Name	Data Type	Description
id	bigint	Unique identifier of the failed background job.
uuid	varchar(255)	Unique job identifier.
connection	text	Queue connection used by the job.
queue	text	Queue where the job was processed.
payload	text	Serialized job data.
exception	text	Exception generated by the failed job.
failed_at	timestamp	Date and time the job failed.

Table 17 Migrations

migrations

Tracks the database migrations applied by Laravel.

Field Name	Data Type	Description
id	integer	Unique migration record identifier.
migration	varchar(255)	Name of the executed migration.
batch	integer	Migration execution batch number.